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Project Title: Model BETA13

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Batch : ECE A

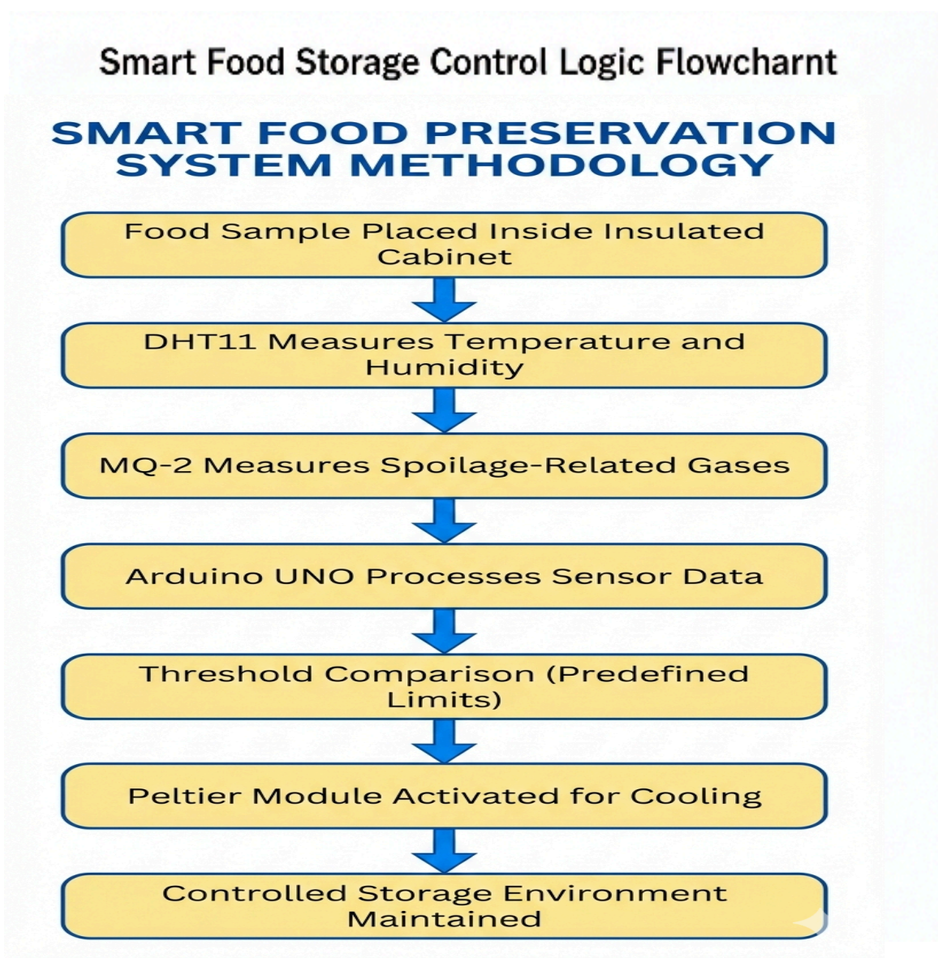
Introduction

This project presents a **smart food preservation system** using **thermoelectric cooling** and **environmental sensing** to monitor and control storage conditions in real time. The system integrates **Arduino UNO** , **Peltier cooling**, and gas-based spoilage detection to provide a low-cost, efficient, and scalable solution.

Objectives

- To design a compact food preservation system using thermoelectric cooling
- To monitor temperature, humidity, and spoilage gases in real time
- To implement monitoring using Arduino UNO To reduce food spoilage through early detection and controlled cooling
- To develop a low-cost and energy-efficient prototype

Methodology



Empirical, Calculations, Tools & Techniques

- Sensors were calibrated using ambient reference readings to ensure measurement accuracy.
- Safe threshold values for temperature, humidity, and gas concentration were defined based on food storage standards.
- Peltier module power requirements were estimated using SMPS voltage and current ratings.
- Cooling performance was evaluated by observing temperature reduction over time inside the insulated chamber.
- MQ2 sensor readings were analyzed to identify spoilage-related gas trends.
- Arduino UNO firmware implemented threshold-based control for automated cooling.
- Tools used include Arduino IDE, UNO .

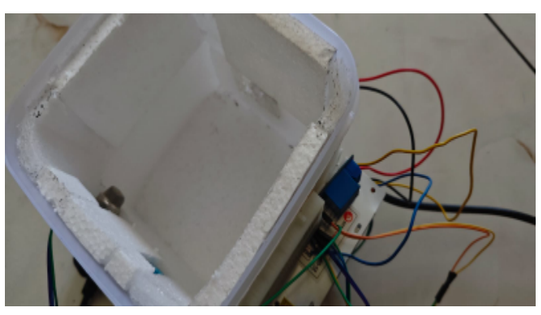
Results & Discussions

- The system successfully monitored temperature, humidity, and gas concentration in real time.
- Activation of the Peltier module resulted in a noticeable and stable reduction in internal temperature.
- MQ2 sensor readings showed a gradual increase in gas concentration with food degradation, validating its use for spoilage indication.
- The insulated chamber minimized thermal losses and improved cooling efficiency.
- Sensor data can be transmitted to the IoT dashboard with minimal delay.
- Overall performance confirms the feasibility of a food preservation system.
- The results validate the feasibility of a sensor-based food preservation prototype with scope for future IoT integration.

Testing/Validation & Feedback

- Functional testing of each sensor independently
- Cooling response verified against temperature rise
- Gas sensor response tested using controlled exposure





Measurable Outcomes

- Internal temperature reduction of approximately **6–10°C** achieved using the Peltier module.
- Stable temperature maintenance observed over extended operating periods.
- Accurate measurement of **temperature (±0.5°C)** and **humidity (±2–5%)** using the DHT11 sensor.
- Detectable increase in gas concentration corresponding to food spoilage using the MQ135 sensor.
- Effective thermal insulation resulting in reduced heat loss from the cabinet.

References

[1] J. W. Grate and A. R. McGill,
“The role of chemical sensors in food quality control,”
IEEE Sensors Journal, vol. 2, no. 6, pp. 576–584, and others

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